

Trainings ▾

Disassemble

⚠ WARNING ⚠

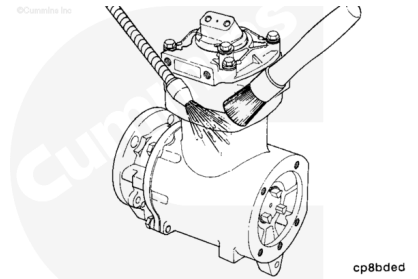
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

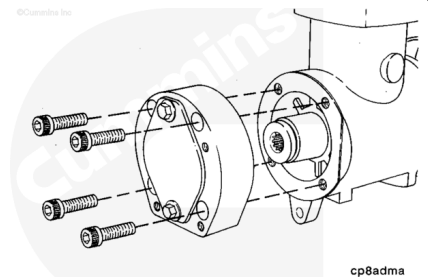
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to clean the air compressor.

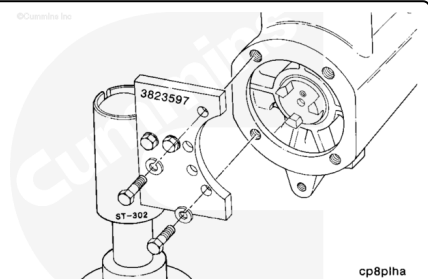
Dry air compressor with compressed air.



Remove the hydraulic pump adapter from the compressor crankcase.



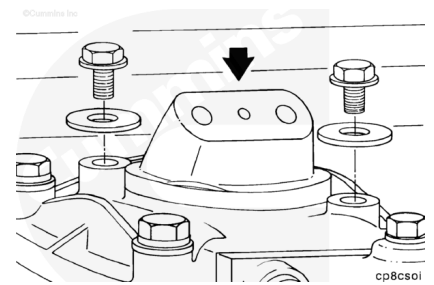
Install the compressor to the mounting plate, Part Number 3823597, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



Disassemble the air compressor cylinder head. For SS and E-Type Models, refer to Procedure 012-103 (</qs3/pubsys2/xml/en/procedures/04/04-012-103.html>). For

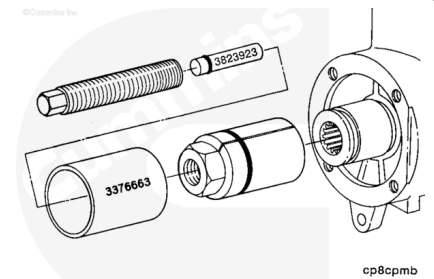
QE models, refer to Procedure 012-104

(/qs3/pubsys2/xml/en/procedures/04/04-012-104.html).

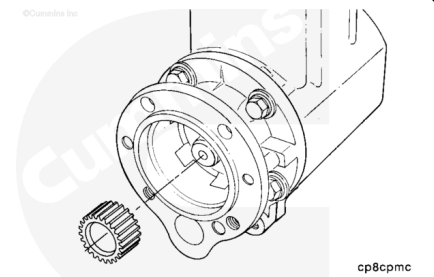


Note : This coupling hub **only** needs to be removed from the crankshaft it is to be replaced.

Use coupling puller, Part Number 3376663, or equivalent, and spacer, Part Number 3823923, or equivalent, to remove the hydraulic pump drive coupling from the crankshaft.

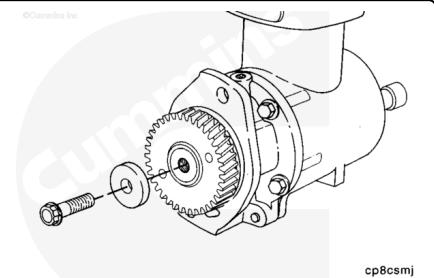


Use a coupling puller, Part Number 3376663, or equivalent, to remove the spline coupling hub.

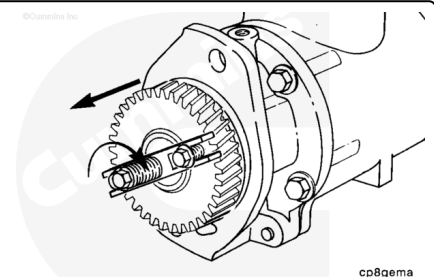


Note : Use a hammer handle or socket drive extension wedged between the crankshaft counterweight and crankcase to keep the crankshaft from turning.

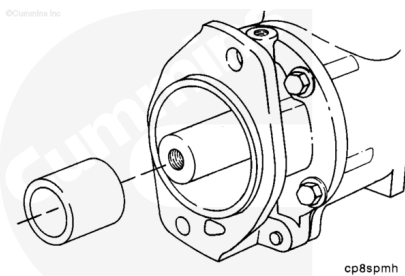
Remove the drive gear retaining capscrew and washer.



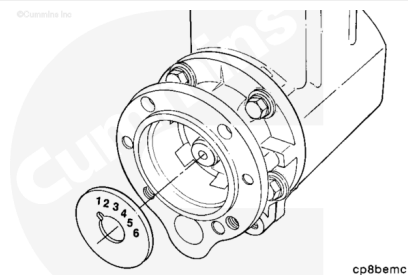
Use puller, Part Number ST-647, or equivalent, to remove the drive gear from the crankshaft. Puller capscrews **must** be M8 x 1.25-6H thread.



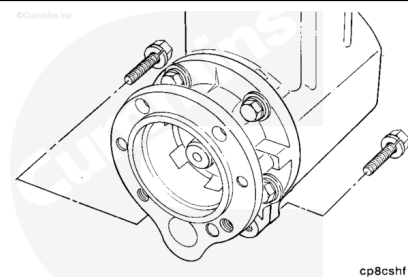
Remove the thrust spacer.



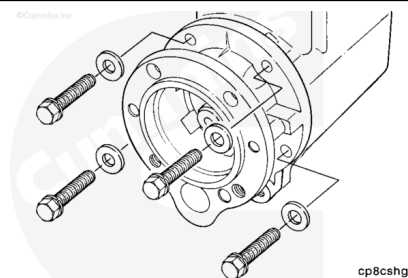
Remove the thrust bearing.



Remove the two hexagon head capscrews and lockwashers.

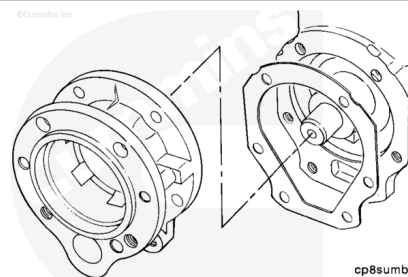


Remove the four captive washer capscrews and the four plain washers.



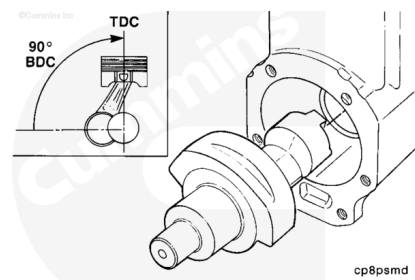
Remove the air compressor support.

Remove and discard the crankcase gasket.

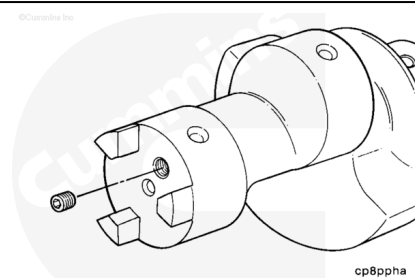


Remove the crankshaft by rotating the crankshaft so that the piston is approximately 90 degrees before or after top dead center (TDC). The piston and crankshaft **must** be in this

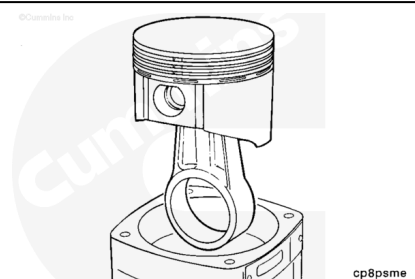
position before the crankshaft can be removed.



Remove the pipe plug.

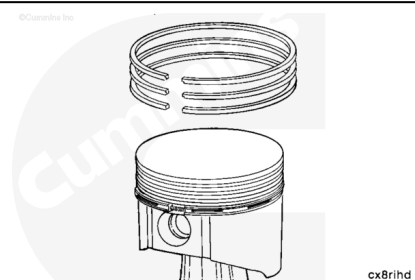


Remove the piston and connection rod assembly.

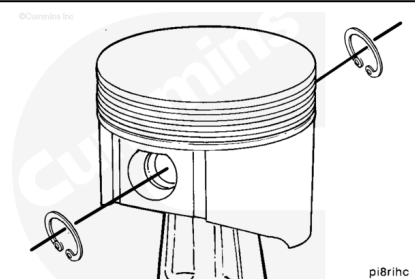


Install the connecting rod in soft jawed vise.

Use a piston ring expander and remove the piston rings.



Use a snap ring pliers and remove the two retaining rings.

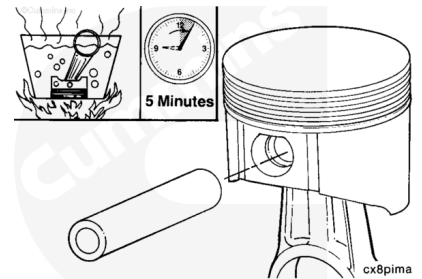


⚠ WARNING ⚠

Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation, or other personal injury or death.

⚠ CAUTION ⚠

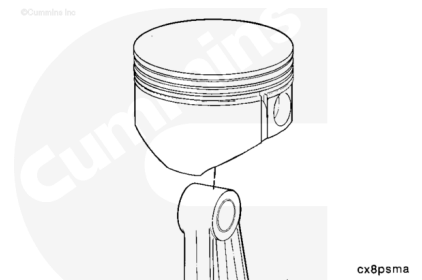
Do not force the piston from the piston or piston damage can occur.



If the pin can **not** be removed by hand pressure, place the piston in boiling water for 5 minutes to expand the pin bore. This will allow the pin to be removed.

Remove the piston pin.

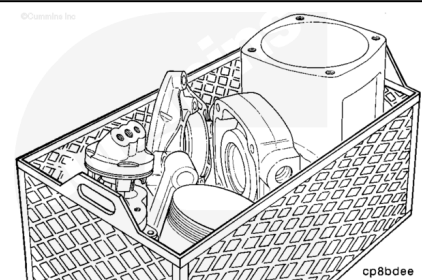
Remove the piston.



Clean and Inspect for Reuse

⚠ WARNING ⚠

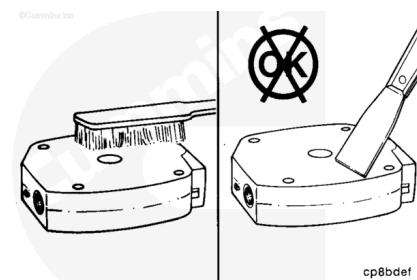
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Soak the parts in kerosene emulsion base cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

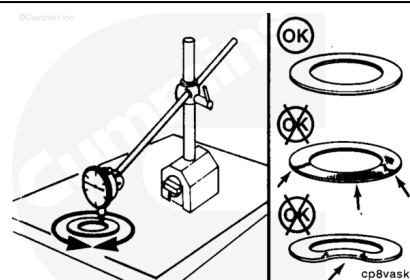
⚠ CAUTION ⚠

Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.



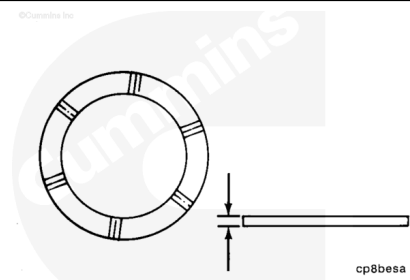
The parts can be scrubbed with a stiff non-metallic bristle brush.

Inspect the cylinder head components. For SS and E-Type models, refer to Procedure 012-103 (</qs3/pubsys2/xml/en/procedures/04/04-012-103.html>). For QE models, refer to Procedure 012-104 (</qs3/pubsys2/xml/en/procedures/04/04-012-104.html>).



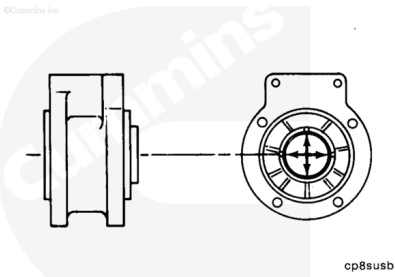
Measure the thrust bearing thickness.

	mm		in
130080	2.24	MIN	0.088
	2.29	MAX	0.090
130081	2.29	MIN	0.090
	2.34	MAX	0.092
130082	2.54	MIN	0.100
	2.59	MAX	0.102
130083	2.34	MIN	0.092
	2.39	MAX	0.094
188040	2.29	MIN	0.090
	2.31	MAX	0.091
188042	2.31	MIN	0.091
	2.34	MAX	0.092
188044	2.34	MIN	0.092
	2.36	MAX	0.093
211662	6.10	MIN	0.240
	6.30	MAX	0.248
3050924	6.10	MIN	0.240
	6.30	MAX	0.248



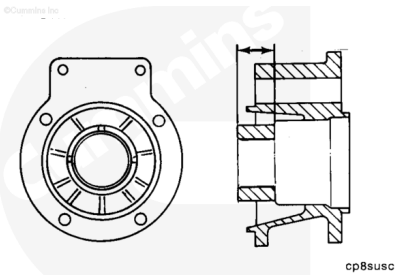
Measure the inside diameter (ID) of the air compressor support.

mm		in
47.600	MIN	1.8740
47.688	MAX	1.8775

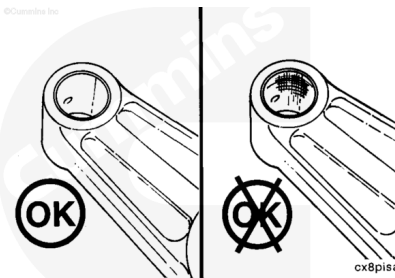


Measure the dimensions between the thrust faces.

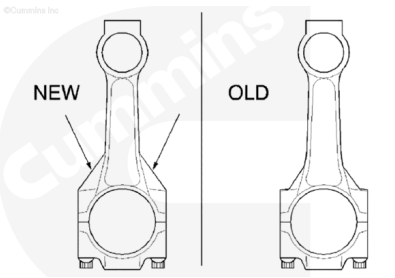
Replace, if worn beyond the specifications given in the Distance Between Thrust Faces Specifications Chart in Section V. Refer to Procedure 018-022 (</qs3/pubsys2/xml/en/procedures/04/04-018-022.html>).



Inspect the connecting rod piston pin bore for damage.
Replace, if damaged.



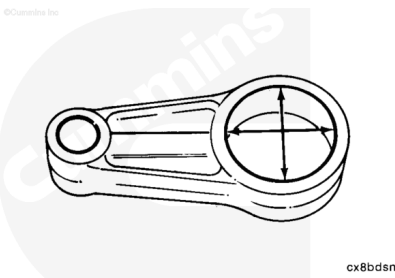
Inspect the connecting rod crankshaft bore for damage.
Replace, if damaged.



Measure the inside diameter of the crankshaft end of the connecting rod.

The inside diameter **must not** to exceed the following:

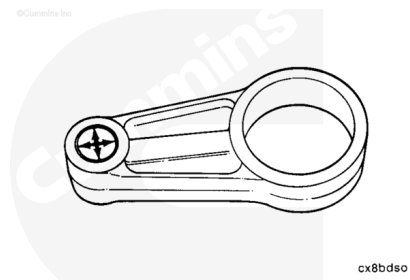
Measurements		
	mm	in
Inside Diameter of Crankshaft Bore	49.263	1.9395



Measure the inside diameter of the piston end of the connecting rod.

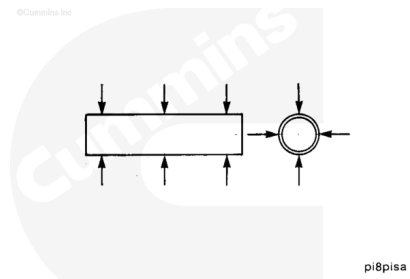
The inside diameter **must not** to exceed the following:

Measurements		
	mm	in
Inside Diameter of Piston Bore	17.513	0.6895



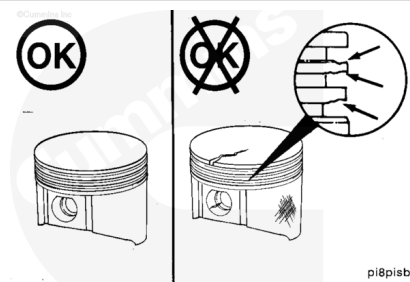
Measure the outside diameter of the piston pin.

mm		in
17.455	MIN	0.6872
17.465	MAX	0.6876



Inspect the piston top and pin bore for cracks.

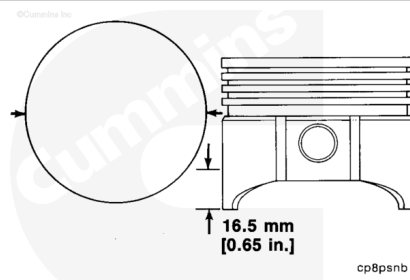
Inspect the ring grooves and skirt for damage. Replace, if damaged.



Measure the outside diameter of the piston.

Measure at 6.35 mm [0.25 in] for the QE230, SS296, SS296E, and QE296 models or measure at 16.5 mm [0.65 in] for the SS338E or QE338 models above the bottom of the piston skirt and at right angle to the piston pin bore.

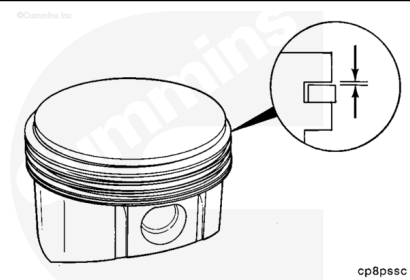
Replace, if worn smaller than 91.87 mm [3.617 in] for QE230, SS296, SS296E, and QE296 models or 98.22 mm [3.867 in] for SS338E or QE338 models.



Use new piston rings and feeler gauge to measure for ring groove wear.

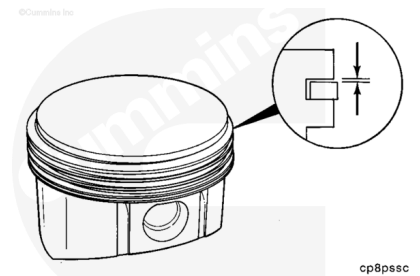
Replace the piston, if the gaps are worn larger than the following dimensions:

	mm		in
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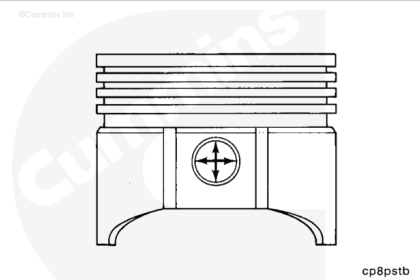
Piston Ring (top)	0.05	MIN	0.002
	0.11	MAX	0.005
Piston Ring (second)	0.05	MIN	0.002
	0.11	MAX	0.005
Piston Ring (oil)	0.05	MIN	0.002
	0.10	MAX	0.004

	mm		in
Piston Ring (top)	0.02	MIN	0.001
	0.08	MAX	0.003
Piston Ring (second)	0.02	MIN	0.001
	0.08	MAX	0.003
Piston Ring (third)	0.02	MIN	0.001
	0.08	MAX	0.003
Piston Ring (oil)	0.03	MIN	0.0012
	0.10	MAX	0.004



Measure the piston pin bore.

Replace the piston, if the bore is worn larger than 17.49 mm [0.689 in].



Inspect the cylinder bore of the crankcase for scoring or other damage.

If the cylinder bore is damaged, bore or hone the cylinder for oversized piston and rings.

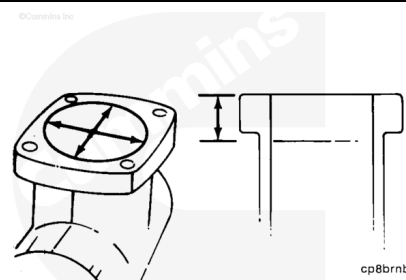


Use a dial bore gauge, Part Number 3375072, or equivalent, to measure the cylinder bore diameter.

Measure at 25.0 mm [1.00 in] below the top of the crankcase.

Maximum out-of-round specification is 0.038 mm [0.0015 in] for the QE230, SS296, SS296E, and QE296.

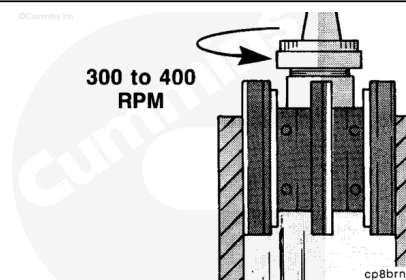
Maximum out-of-round specification is 0.05 mm [0.002 in] for the SS338E and QE338.



	mm		in
SS338E and QE338	98.43	MIN	3.875
	98.51	MAX	3.879
QE230, SS296, SS296E, and QE296	92.08	MIN	3.625
	92.16	MAX	3.628

If the crankcase bore is **not** within specifications, bore the cylinder to use oversize piston and rings.

Bore or hone the cylinder diameter (per the following instructions) to the oversize specifications listed in the following chart:

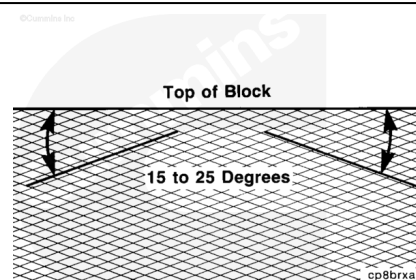


Cylinder Bore			
		Diameter mm [in]	
	Oversize mm [in]	Minimum	Maximum
SS338 and QE338	0.25 [0.010]	98.68 [3.885]	98.72 [3.887]
	0.51 [0.020]	98.93 [3.895]	98.97 [3.897]
	0.76 [0.030]	98.19 [3.905]	98.23 [3.907]

Cylinder Bore

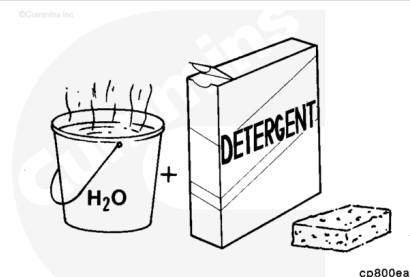
		Diameter mm [in]	
QE230, SS296, SS296E, and QE296	0.25 [0.010]	92.33 [3.635]	92.36 [3.6365]
	0.51 [0.020]	92.58 [3.645]	92.621 [3.6465]
	0.76 [0.030]	92.84 [3.655]	92.875 [3.6565]

Use a 280 grit hone to produce a cross hatched surface finished with lines at a 15 to 25 degree angle with the top of the crankcase.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use a strong solution of laundry detergent and hot water to clean the cylinder bore.

Dry with compressed air.

Use a white lint-free, lightly oiled cloth to check the bore for cleanliness.

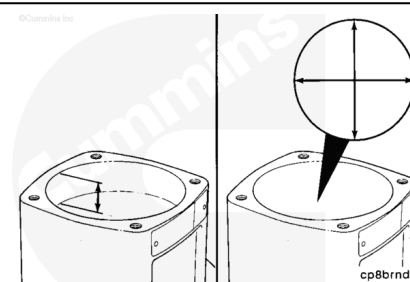
If grit is still present, clean cylinder bore again.



Use a dial bore gauge, Part Number 3375072, or equivalent, to measure the cylinder bore diameter.

Measure at 25.0 mm [1.00 in] below the top of the crankcase.

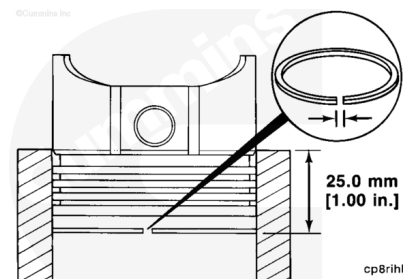
Maximum out-of-round is 0.04 mm [0.002 in].



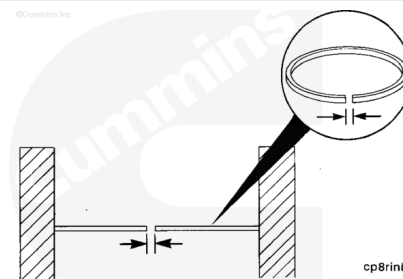
Note : Cummins Inc. recommends that new piston rings be installed during rebuild.

If old piston rings are used, follow the instructions listed below to measure the ring gaps.

Insert one ring at a time into the cylinder bore. Seat the ring with a piston head squarely 25.0 mm [1.00 in] below the top of the crankcase.

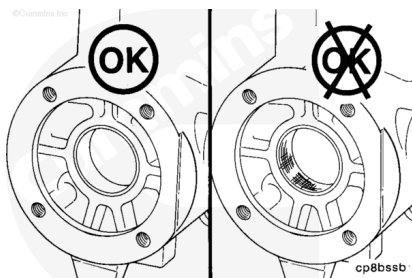


	mm		in
Piston Ring (top)	0.40	MIN	0.016
	0.65	MAX	0.026
Piston Ring (second)	0.40	MIN	0.016
	0.65	MAX	0.026
Piston Ring (third)	0.40	MIN	0.016
	0.65	MAX	0.026
Piston Ring (oil)	0.30	MIN	0.012
	0.60	MAX	0.024



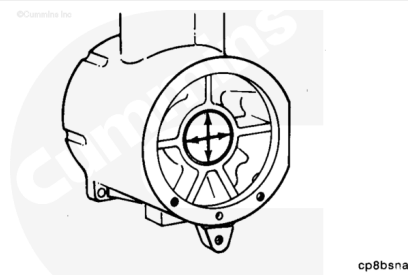
	mm		in
Piston Ring (top)	0.25	MIN	0.010
	0.51	MAX	0.020
Piston Ring (second)	0.25	MIN	0.010
	0.51	MAX	0.020
Piston Ring (oil)	0.38	MIN	0.015
	1.40	MAX	0.055

Inspect the bushing in the crankcase and replace, if damaged.

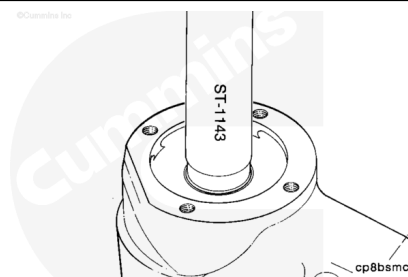


Measure the inside diameter.

Replace the bushing if worn larger than 47.70 mm [1.8780 in].



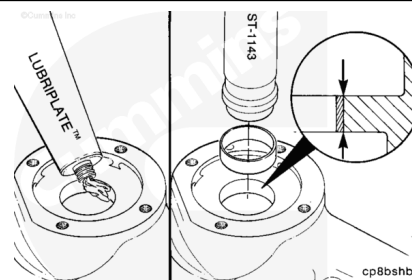
Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to remove the bushing.



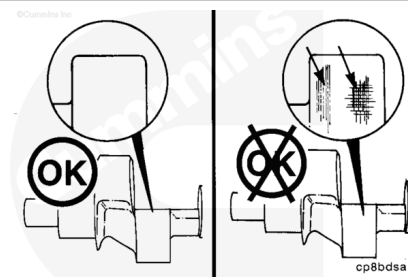
Use clean Lubriplate™ to lubricate the bushing bore in the crankcase.

Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to install the new bushing.

Push in the new bushing until it is flush with the bore surface. Oil holes in the bushing and crankcase **must** be aligned.

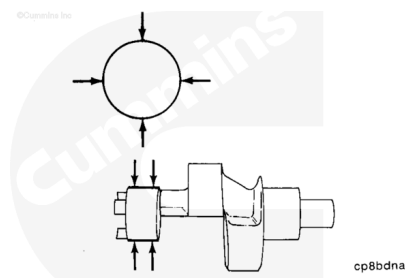


Inspect the crankshaft for scratches and scoring. Replace, if damaged.



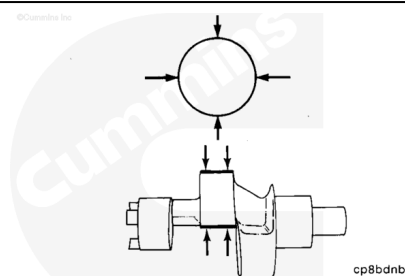
Measure the outside diameter of the journal at the crankcase (A).

Replace if worn beyond specifications given in the Crankshaft Dimensions Chart in Section V. Refer to Procedure 018-022 (</qs3/pubsys2/xml/en/procedures/04/04-018-022.html>).



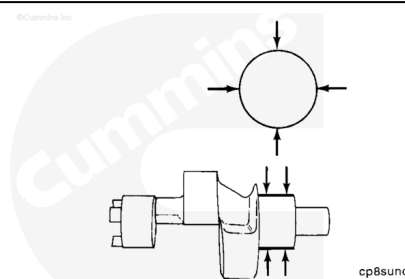
Measure the outside diameter of the journal at the connecting rod (B).

Replace if worn beyond specifications given in the Crankshaft Dimensions Chart in Section V. Refer to Procedure 018-022 (</qs3/pubsys2/xml/en/procedures/04/04-018-022.html>).



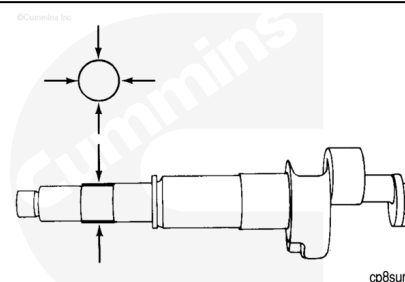
Measure the outside diameter of the journal at the air compressor support (C).

Replace if worn beyond specifications given in the Crankshaft Dimensions Chart in Section V. Refer to Procedure 018-022 (</qs3/pubsys2/xml/en/procedures/04/04-018-022.html>).

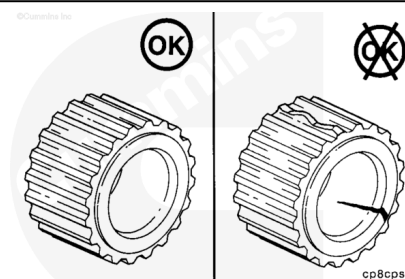


For SS296, SS296E, and QE296, measure the outside diameter of the journal at the engine gear case journal (long nose crankshaft) (D).

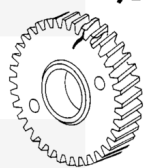
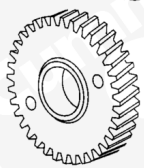
Replace if worn beyond specifications given in the Crankshaft Dimensions Chart in Section V. Refer to Procedure 018-022 (</qs3/pubsys2/xml/en/procedures/04/04-018-022.html>).



Inspect the spline coupling hub for cracks, worn, or broken teeth. Replace, if damaged.



Inspect the drive gear for cracks (gear driven compressors **only**), worn or broken teeth. Replace, if damaged.

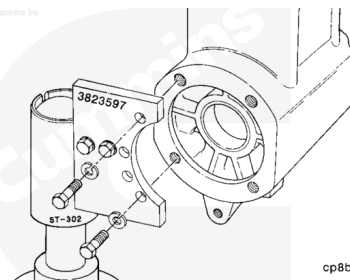


cp8gesa

Assemble

Install the crankcase to the mounting plate, Part Number 3823597, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.

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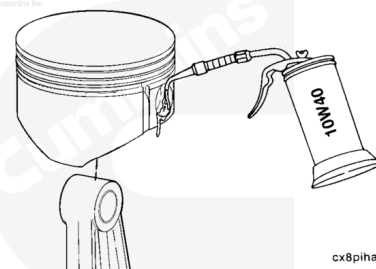


cp8bdhd

Install the connection rod in a soft jawed vise.

Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore.

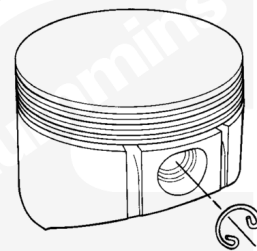
©Cummins Inc.



cx8piha

Install the retaining ring.

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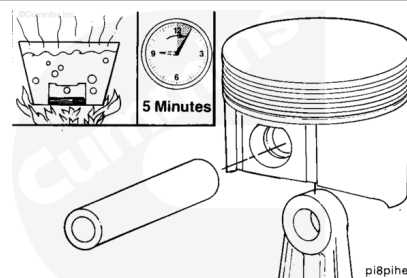


pi8rihb

⚠ WARNING ⚠

Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation, or other personal injury or death.

⚠ CAUTION ⚠



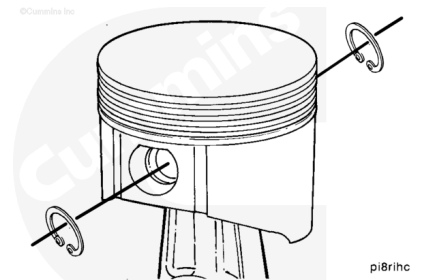
pi8pihe

Do not force the piston from the piston or piston damage can occur.

If the pin can **not** be removed by hand pressure, place the piston in boiling water for 5 minutes to expand the pin bore. This will allow the pin to be removed.

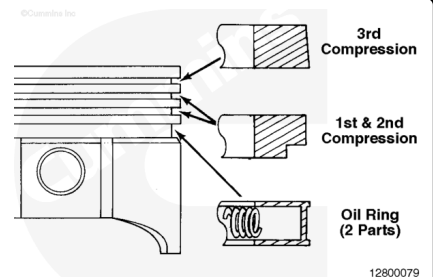
Remove the piston pin.

Install the second retaining ring.



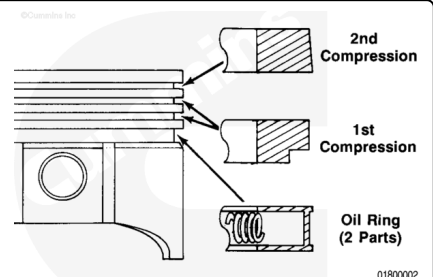
Note : The first and second compression rings are hook scraper, and the third compression ring is a taper face.

Piston ring positions for SS338 and QE338.



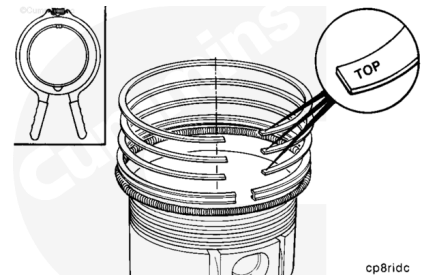
Note : The first and second compression rings are hook scraper, and the third compression ring is a taper face.

Piston ring positions for QE230, SS296, SS296E, and QE296.

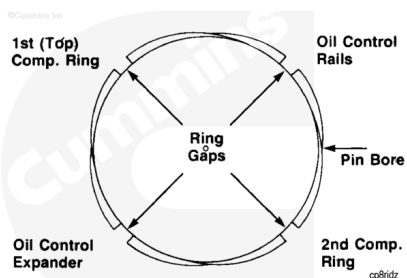


Note : Cummins Inc. recommends that new rings be installed during rebuild.

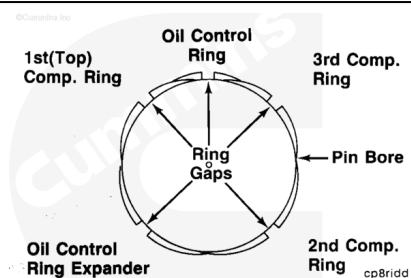
Install the piston rings with the part number or side marked "TOP" up.



Position the QE230, SS296, SS296E, and QE338 piston ring gaps so they are **not** over the piston pin bore.

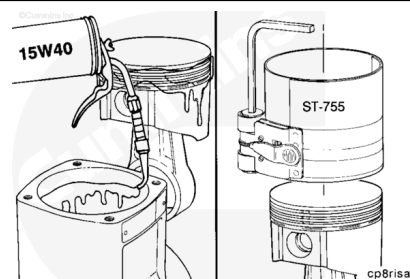


Position the SS338E and QE338 piston ring gaps so they are **not** over the piston pin bore.

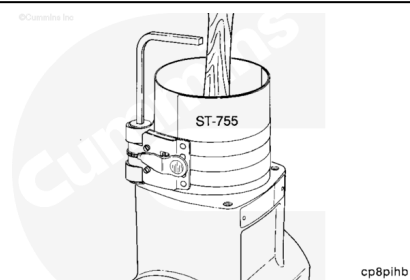


Use clean 15W-40 lubricating engine oil to lubricate the rings, piston, and crankcase.

Use a piston ring compressor, Part Number ST-755, or equivalent, to compress the piston rings.

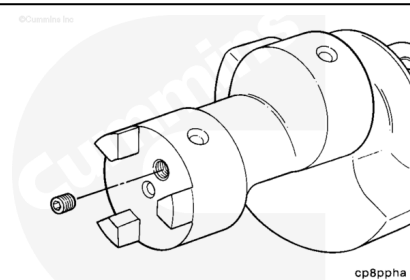


Install the piston and connecting rod assembly.



Install and tighten the pipe plug.

Torque Value: 8 n•m [71 in-lb]



⚠ CAUTION ⚠

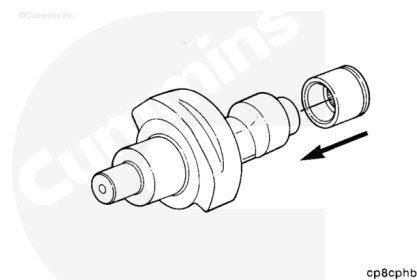
Excessive load from the press can bend the crankshaft.

Note : Do not exceed 53379 N [12,000 lbf] on press.

This step applies to gear drive air compressors **only**.

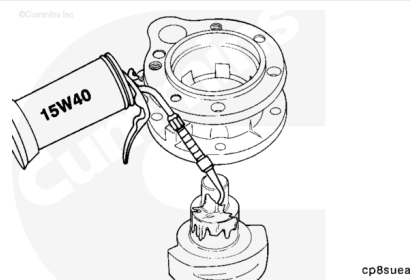
Use a press to install the hydraulic pump drive coupling onto the crankshaft until it bottoms.

Note : The free end of the crankshaft must be supported when installing the coupling. Couplings can not be installed with the air compressor on the engine.



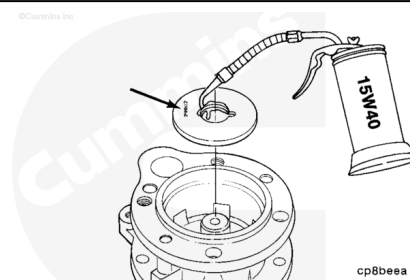
Use clean 15W-40 lubricating engine oil to lubricate the crankshaft and support bore.

Install the support onto the crankshaft.



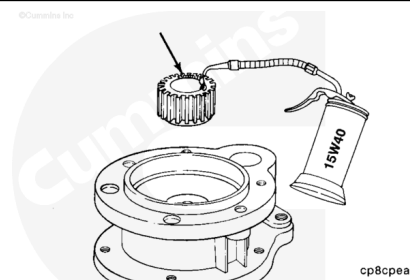
Use clean 15W-40 lubricating engine oil to lubricate the thrust bearing.

Install the thrust bearing with the part number side "OUT" or the grooved side "IN".



Use clean 15W-40 lubricating engine oil to lubricate the spline coupling hub bore.

Position the spline coupling hub with the flanged side "IN" or the part number side "OUT".

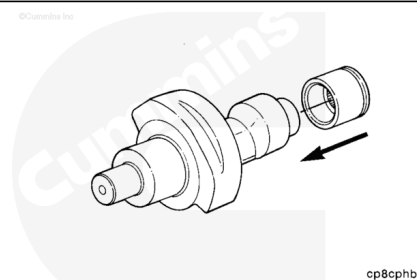


⚠ CAUTION ⚠

Excessive load from the press can bend the crankshaft.

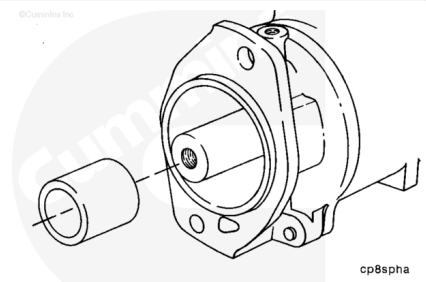
Note : Do not exceed 53379 N [12,000 lbf] on press.

Use a hand or hydraulic press to push the spline coupling hub into the crankshaft until it contacts the thrust bearing.



Note : The free end of the crankshaft must be supported when installing the coupling. Couplings can not be installed with the air compressor on the engine.

Install the thrust spacer.



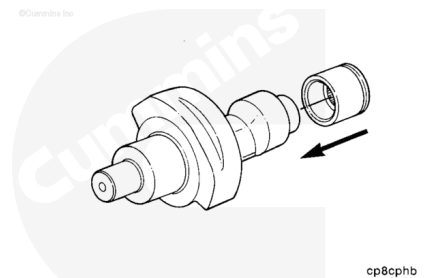
⚠ CAUTION ⚠

Excessive load from the press can bend the crankshaft.

Note : Do not exceed 53379 N [12,000 lbf] on press.

Use a press to push the drive gear onto the crankshaft with the part number side "OUT" until the back side of the gear bottoms against the crankshaft shoulder.

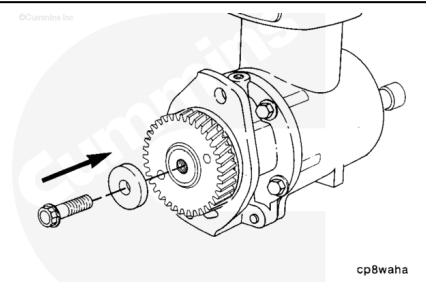
Note : The free end of the crankshaft must be supported when installing the coupling. Couplings can not be installed with the air compressor on the engine.



This step applies to gear driven air compressors **only**.

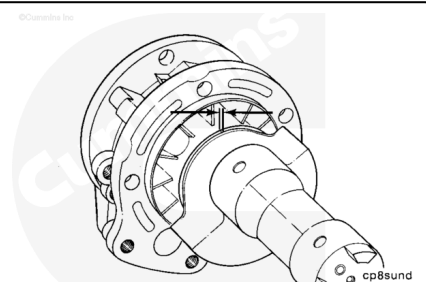
Install the washer and drive gear retaining capscrew.

Torque Value: 100 n•m [75 ft-lb]



Measure the clearance between the support and the crankshaft mating surfaces.

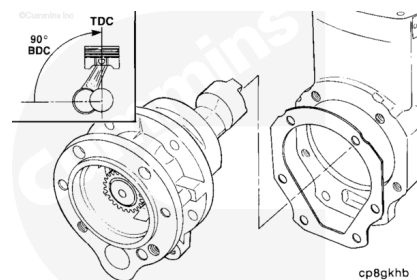
Reference the Crankshaft End Clearance chart in Section V.
Refer to Procedure 018-022
(/qs3/pubsys2/xml/en/procedures/04/04-018-022.html).



Position the piston at 90 degrees after top dead center (TDC). This will allow the crankshaft to be installed through the connecting rod.

Install a new gasket on the support.

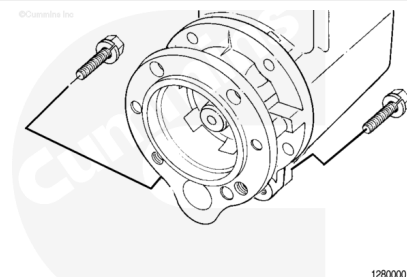
Install the crankshaft.



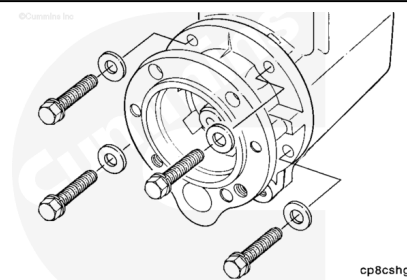
⚠ CAUTION ⚠

Do not use the capscrews to pull the supporting housing crankshaft assembly into the crankcase. Damage to the connecting rod can result.

Install the two captive washer capscrews.

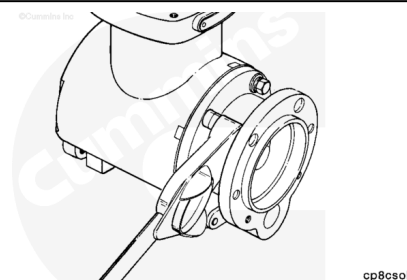


Install the four flat washers and the four captive washer capscrews.



Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]

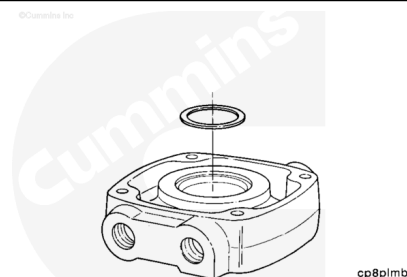


Assemble the cylinder head. For SS and E-Type models, refer to Procedure 012-103

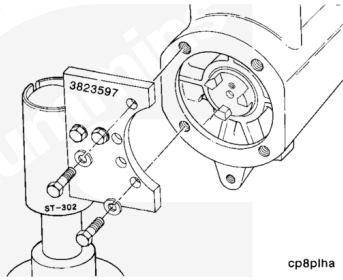
(</qs3/pubsys2/xml/en/procedures/04/04-012-103.html>).

For QE models, refer to Procedure 012-104

(</qs3/pubsys2/xml/en/procedures/04/04-012-104.html>).



Remove the compressor from the ball joint vise and mounting plate.



This step applies to gear driven air compressors **only**.

Install the hydraulic pump adapter and capscrews.

Torque Value: 60 n•m [45 ft-lb]

